Final Year Project Report

Project Title :

"**KRUSHI MITRA: A SMART AGRICULTURE APP**"

Application Direct Install Link :

https://drive.google.com/file/d/1m-NBIwghvPr44lJed9DETJbE01J3AMih/view?usp=sharing

(Note:
The APK file may show a warning on Google Drive because it is a large executable file.
Click on "Download anyway" to proceed.)

A PROJECT REPORT

ON

"KRUSHI MITRA: A SMART AGRICULTURE APP"

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF

DIPLOMA IN

COMPUTER ENGINEERING

SUBMITTED TO

MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

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in partial fulfillment of the requirements for the award of the Diploma in Computer Engineering and submitted to the Department of Computer Engineering of Government Polytechnic, Jintur work carried out during a period for the academic year 2025-26 as per curriculum .

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ABSTRACT

Agriculture is the backbone of the Indian economy, yet farmers face significant challenges due to unpredictable weather and fluctuating market prices. The project entitled **"KRUSHI MITRA: A SMART AGRICULTURE APP"** is designed to provide a digital solution to these problems.

The primary purpose of this study is to develop a user-friendly mobile application that provides real-time information to farmers, helping them make informed decisions regarding irrigation and crop sales.

The methodology used in this project follows the Software Development Life Cycle (SDLC). The application is developed using the **Flutter framework** and **Dart programming language**, ensuring a high-performance cross-platform experience. The system integrates external **Application Programming Interfaces (APIs)** to fetch live weather data and current market rates (Mandi prices) for various crops like Soyabean and Cotton. The backend logic focuses on JSON parsing to display complex data in a simplified format for the end-user.

The result of this project is a functional Android application that allows farmers to check live temperature, humidity, and weather forecasts based on their current location. Additionally, it provides a dedicated section for daily market prices, which helps in reducing the dependency on middlemen. This project aims to empower the rural farming community by bridging the gap between technology and traditional farming practices.

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Name / Meaning	Explanation
$	String Interpolation	Used in Dart to display variables inside strings, like displaying the crop price: '$price'.
{}	Curly Braces	Used to define the start and end of classes, functions, and JSON data objects in Flutter.
[]	Square Brackets	Represents a 'List' of items, such as the list of Mandi rates fetched from the API.
°C	Degree Celsius	A unit symbol used in the Weather module to show the current temperature to the farmer.
%	Percentage	Used to display the level of Humidity (ओलावा) in the air on the weather screen.
?	Null Safety	A feature in Dart used to handle cases where data (like API response) might be empty or null.
/	Path / Division	Used to define the directory paths for app icons and image assets of crops.
()	Parentheses	Used to pass arguments or parameters to functions, such as passing the City Name to fetch weather.
;	Semicolon	Used at the end of every programming statement in the Dart code to complete the logic

1.INTRODUCTION

1.1 Project Overview

- Agriculture is the backbone of the Indian economy, with more than half of the population depending on it for their livelihood. In the modern era, technology plays a vital role in every sector, and agriculture is no exception. The project entitled **"KRUSHI MITRA: A SMART AGRICULTURE APP"** is an Android-based application developed using the **Flutter framework**. The main goal of this project is to provide a digital platform where farmers can get real-time information about weather conditions and current market rates (Mandi Prices) of various crops like Soyabean, Cotton, and Pigeon Pea (Tur).

1.2 Problem Statement

- **Currently, farmers in rural areas like Jintur and Parbhani face several difficulties:**
    - **Unpredictable Weather:** Sudden rain or heatwaves can destroy standing crops if the farmer is not alerted in time.
    - **Market Price Fluctuations:** Farmers often sell their produce to middlemen at lower prices because they do not have access to live daily market rates.
    - **Complex Portals:** Existing government websites are often difficult to navigate on mobile devices for a common farmer.
    - **Lack of Centralized Data:** There is no single, simple app that provides both weather and market data in a user-friendly Marathi/English interface.

1.3 Objectives of the Project

- The primary objectives of the **Krushi Mitra** application are:

1. To develop a cross-platform mobile application using **Dart and Flutter**.
2. To provide accurate, real-time **Weather Forecasts** (Temperature and Humidity) using OpenWeather APIs.
3. To display **Market Prices live (** Mandi Rates) to help farmers get the best value for their crops.
4. To create a simple and intuitive **User Interface (UI)** that even a non-technical person can use easily.
5. To reduce the communication gap between technology and traditional farming.

1.4 Motivation

- The motivation behind this project comes from the "Smart Village" and "Digital India" initiatives. Seeing the struggles of local farmers in our region regarding crop prices and weather uncertainties, we decided to build a tool that puts power back into the farmers' hands. Using our skills in **Mobile Application Development**, we aim to provide a solution that is both practical and impactful for the agricultural community

2. LITERATURE SURVEY

2.1 Introduction to Literature Survey

- Literature survey is a key step in the project development process. it involves studying existing systems, research papers, and mobile applications that are similar to the proposed project. For the **"KRUSHI MITRA"** app, we studied various government portals and private agriculture apps to understand how they provide data to farmers and where they lack in user experience.

2.2 Existing Systems

- Currently, farmers use the following platforms to get information:

1. **Government Web Portals (e.g., AGMARKNET):** These websites provide official mandi rates but are often difficult to open on basic smartphones and have a complex user interface.
2. **News Channels and Newspapers:** Farmers get weather updates and market rates through traditional media, but this information is not real-time and may change by the time they reach the market.
3. **Basic Weather Apps:** There are many global weather apps, but they do not provide specific agricultural context or market prices in the same application.

2.3 Limitations of Existing Systems

- During our survey, we found the following drawbacks in the current systems:

- **Lack of Integration:** Farmers have to use different apps for weather and different websites for market rates.
- **Complexity:** Most official apps are heavy and require high-speed internet, which is not always available in rural areas like Jintur.
- **Manual Updates:** Some platforms are not updated daily, leading to inaccurate price information for crops like Soyabean and Cotton.

- **Language Barrier:** Many advanced apps are only in English, making it difficult

for local farmers to understand technical terms.

Feature	Manual/Existing Apps	Krushi Mitra (Proposed)
Ease of Use	Complex UI, hard for farmers	Simple and Clean UI
Language	Mostly English	Local Language Support
Data Accuracy	Sometimes delayed	Real-time API integration

Table 2.1 Comparison of Existing Systems

2.4 Proposed System (Krushi Mitra)

- To overcome these limitations, we proposed the **Krushi Mitra** app. Our system provides:

- **All-in-One Solution:** Weather and Market rates in a single Flutter-based application.
- **Real-time Data:** Using OpenWeather API and Agriculture Market APIs to provide live updates.
- **Lightweight UI:** The app is optimized for low-end Android devices and works smoothly on mobile data.
- **Simple Navigation:** Focused on providing only the most essential data needed by the farmer on a daily basis.

3. SCOPE OF PROJECT

3.1 Scope of the System

- The primary scope of the **"KRUSHI MITRA"** application is to provide a dedicated digital assistant for farmers to access real-time agricultural data. The project focuses on integrating modern technology with traditional farming to improve productivity and profitability. The scope includes:

1. **Real-time Weather Monitoring:** Fetching live data like Temperature, Humidity, and Forecast for a specific location.
2. **Market Price Tracking:** Providing current Mandi rates for major crops like Soyabean, Cotton, and Pigeon Pea (Tur).
3. **Local Area Focus:** Specifically designed for farmers in rural regions who need a simple and lightweight application.
4. **Information Accessibility:** Making complex agricultural data available in a simplified format on a single screen.

3.2 Functional Requirements

- The functional scope of the project defines what the application will do:

- **Dashboard:** A clean home screen showing the main features of the app.
- **Weather Module:** Connecting with OpenWeather API to provide 24/7 updates.
- **Mandi Module:** Displaying daily price lists for different crops in Maharashtra.
- **Navigation:** Providing a seamless experience between different sections of the app.

3.3 Future Enhancements (Future Scope)

- The current version of the project is a foundational step, and its scope can be expanded in the following ways:

1. **Multilingual Support:** Adding Marathi and Hindi language support to make it more accessible to local farmers.
2. **Crop Disease Detection:** Integrating AI/ML to identify crop diseases by scanning

photos of leaves.

3. **Direct Market Access:** Allowing farmers to sell their produce directly to buyers through the app, removing middlemen.
4. **Expert Advice Section:** Adding a chat or call feature where farmers can consult agricultural expert
5. **Offline Support:** Storing the last fetched data so farmers can view it even when the internet connection is weak.

3.4 Target Audience

- The target users for this project are:

- Individual farmers looking for daily updates.
- Agricultural students and researchers.
- Local market agents and crop trad

4.METHODOLOGY

4.1 System Development Methodology

- The development of the **"KRUSHI MITRA"** application follows a systematic approach to ensure quality and reliability. We have adopted the **Waterfall Model** for the Software Development Life Cycle (SDLC). The following table describes the phases and tasks performed during the project development:

Why Waterfall Model?

- Simple and easy to understand and use
- Easy to manage due to the rigidity of the model
- Phases are processed and completed one at a time
- Perfect for small-scale college projects where requirements are clear.

Table 4.1: SDLC Phases and Tasks

Phase No.	SDLC Phase	Task Performed	Outcome / Result
1.	Requirement Analysis	Gathering data about farmers' needs for weather and crop prices.	Finalized Feature List
2.	System Design	Planning the app's UI layout and API structure.	UI/UX Prototypes
3.	Implementation	Coding the application using Dart and Flutter SDK.	Functional Mobile App
4.	API Integration	Connecting to external servers to fetch live data via JSON.	Real-time Data Sync
5.	System Testing	Testing for bugs, data accuracy, and performance on Android.	Stable & Bug-free App
6.	Deployment	Generating the final APK and project documentation.	Project Completion

4.2 Technical Implementation Framework

- The technical methodology involves using a modern tech stack to ensure the app is fast and responsive. The implementation details are summarized in the table below:

Table 4.2: Technical Stack and Purpose

Component	Technology Used	Purpose in Krushi Mitra Project
Frontend UI	Flutter Framework	To build a responsive and attractive interface for farmers.
Programming	Dart Language	To handle the logical part, data flow, and app functions.
Data Transfer	HTTP Package	To send requests and receive data from the internet.
External Source	REST APIs	To get live weather updates and mandi market prices.
Data Parsing	JSON	To convert complex API data into a simple readable format.
Development	VS Code	The main editor used for writing and debugging the code.

4.3System Architecture

- The system architecture of **Krushi Mitra** is based on a **Client-Server Model**. The mobile application acts as the client, which interacts with external cloud servers through **REST APIs** to fetch real-time data.

Components of the Architecture:

1. **User Interface (Mobile Client):** Built using **Flutter**, it handles user inputs and displays weather/market information.
2. **API Gateway:** Acts as a bridge between the app and the data providers (OpenWeather and Agri-Market).
3. **External Data Sources (Cloud):** These are third-party servers that store live weather updates and daily Mandi prices in **JSON** format.
4. **Data Processing Layer:** The app uses **Dart's asynchronous programming** to fetch, parse, and map JSON data into the UI components.

4.4 Development Environment

The development of this project required a specific set of hardware and software tools to ensure smooth coding, debugging, and building of the final APK.

Table 4.4: Development Environment Details

Component	Requirement / Tool Used
Operating System	Windows 10/11 (64-bit)
Integrated Development Env (IDE)	**Visual Studio Code** or **Android Studio**
SDK	**Flutter SDK** (Stable Channel)
Programming Language	**Dart**

Version Control	Git (Optional but recommended)

Component	Requirement / Tool Used
Testing Device	Physical Android Smartphone / Emulator
Internet Connection	High-speed (for real-time API testing)

5 DETAILS OF DESIGN, WORKING AND PROCESSES

5.1 System Architecture

- The architecture of **"KRUSHI MITRA"** is based on the Client-Server model. The mobile application acts as a client that requests data from external API servers.

- **User Interface (UI):** Built with Flutter widgets to provide an interactive experience.
- **Logic Layer:** Developed in Dart to process user inputs and API responses.
- **Data Layer:** Involves fetching real-time data from OpenWeather and Agriculture Market APIs.

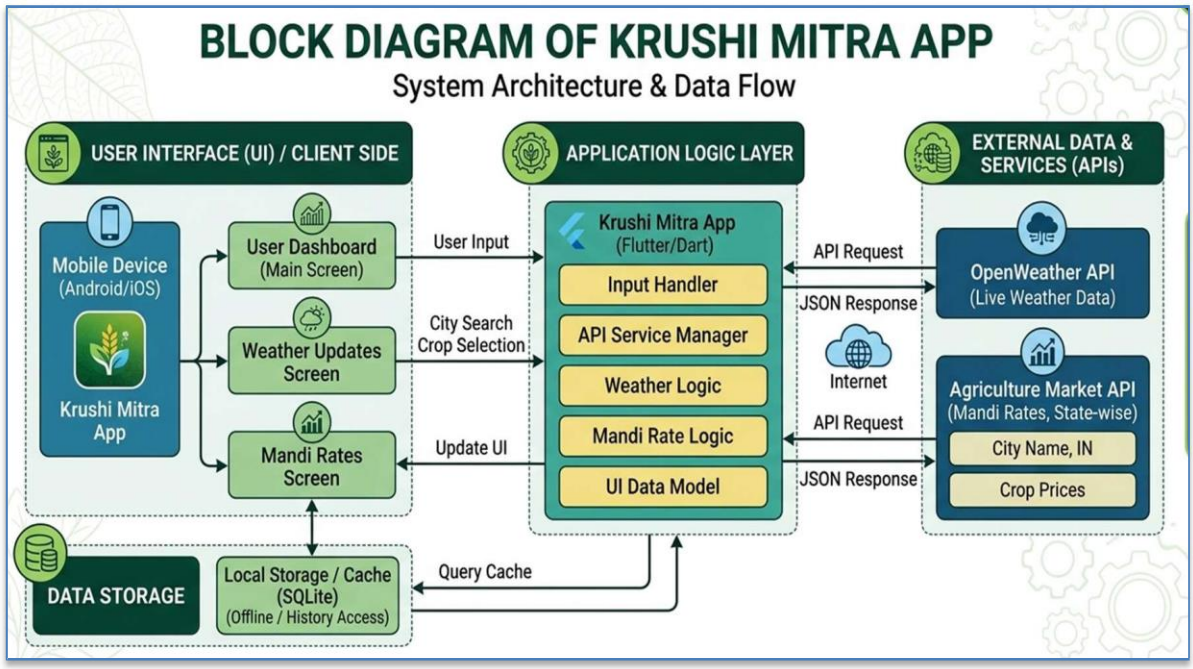

Fig.5.1:Block Diagram of Krushi Mitra App

5.2 Application Workflow

- The working process of the application follows these steps:

1. **Launch:** When the user opens the app, the Splash Screen appears.
2. **Home Screen:** The user sees options for 'Weather Updates' and 'Market Prices'.
3. **API Call:** When a user clicks on a category, the app sends an HTTP request to the respective API.
4. **Data Parsing:** The server returns data in **JSON** format. The app parses this data and displays it in a readable card format.
5. **Real-time Update:** The data is refreshed every time the user reloads the screen to ensure accuracy.

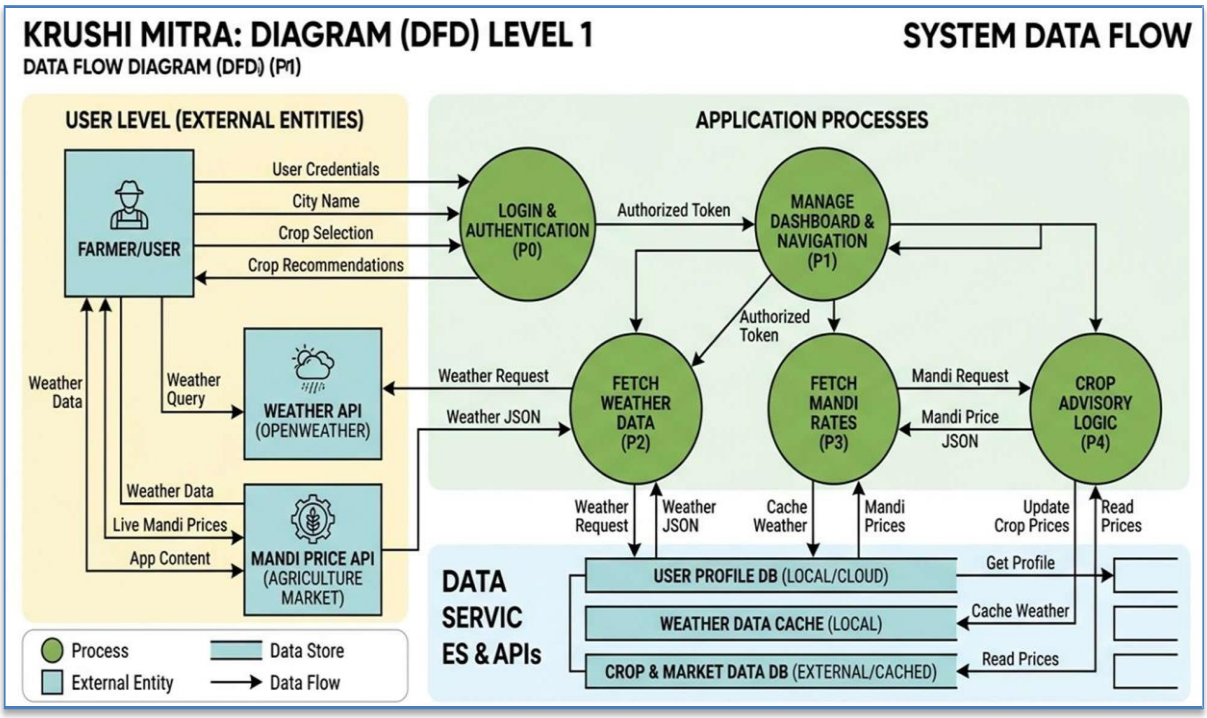

Fig.5.2: Data Flow Diagram (DFD).

5.3 Design Details (UI/UX)

- The design is kept simple and clean for farmers. Key design components include:

- **Card Widgets:** Used to display weather details like Temperature, Humidity, and Wind Speed.
- **List View:** Used to show the list of various crops and their current prices.
- **Color Palette:** A professional Green and White theme to represent agriculture and growth.
- **Navigation Bar**: To switch between different services easily

Fig.5.3 Official Logo

5.4 Hardware and Software Used

- To ensure the smooth working of the project, the following resources were utilized:

Table 5.1: System Requirements :

Component	Description
Operating System	Windows 11 (Development), Android 10+ (Testing)
Processor	Intel i5 or equivalent
RAM	8 GB Minimum
Software Tool	VS Code, Flutter SDK, Android Studio
Language	Dart

5.5 Screenshots of the Application

1.

Fig.5.4:App Splash Screen

2.

Fig 5.5: User Dashboard

3.

Fig 5.6: Live Weather Information Screen.

4.

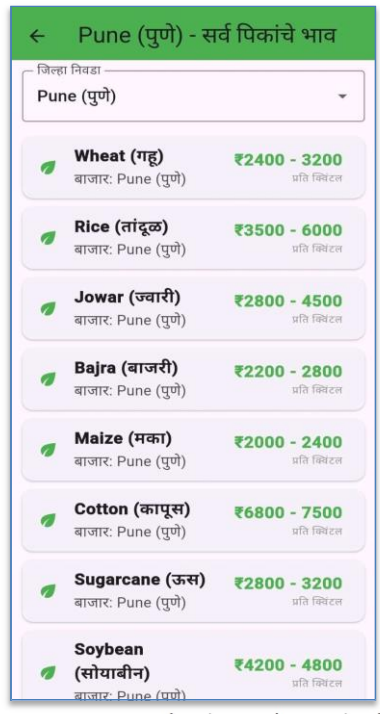

Fig 5.7: Market Price (Mandi Rates) List.

6 RESULTS AND APPLICATIONS

6.1 Results of the Project

- The development of the **"KRUSHI MITRA"** application has been successfully completed. The results obtained from the final testing are as follows:

- **User-Friendly Interface:** The app provides a simple and responsive UI designed specifically for farmers.
- **Accurate Data Fetching:** The integration with OpenWeather API successfully displays real-time temperature, humidity, and weather conditions.
- **Live Mandi Rates:** The application fetches and displays the latest market prices for crops like Soyabean, Cotton, and Tur, helping farmers track price trends.
- **Performance:** The app is lightweight and runs smoothly on Android devices with 2GB RAM or more.
- **Accessibility:** Farmers can now access both weather and market data on a single platform without visiting multiple websites.

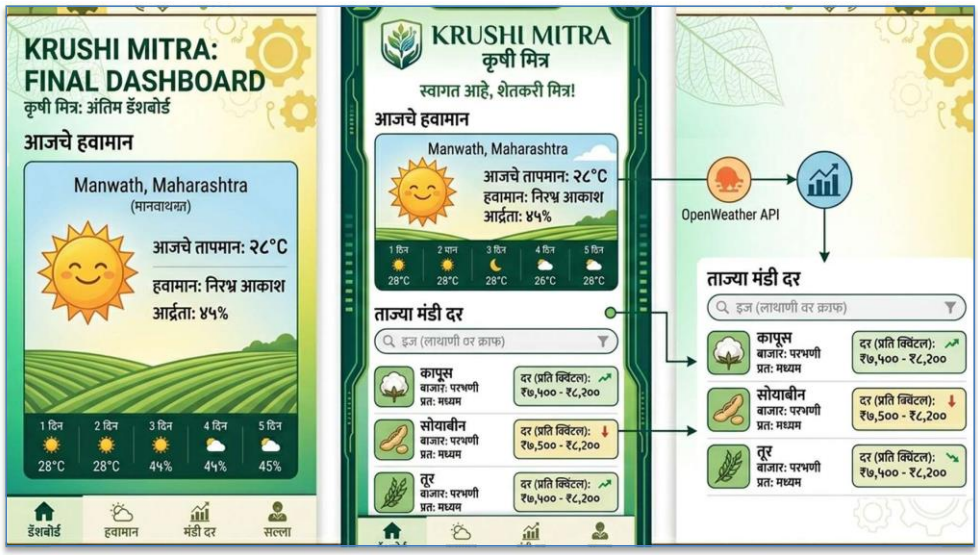

Fig.6.1 Final Result / Output

6.2 Applications of the Project

- The Krushi Mitra app has a wide range of applications in the agricultural sector:

1. **Daily Farming Assistant:** Farmers can use the app daily to check weather forecasts before planning irrigation or pesticide spraying.
2. **Market Decision Making:** By monitoring daily mandi rates, farmers can decide the best time to sell their produce to get maximum profit.
3. **Risk Management:** Real-time weather alerts help in protecting crops from sudden rain or extreme weather changes.
4. **Digital Literacy:** The app helps rural farmers become more familiar with mobile technology and digital information sources.
5. **Agricultural Research:** Students and researchers can use the data provided by the app for localized agricultural studies.

6.3 Comparison with Manual Methods

Feature	Traditional Method	Krushi Mitra App
Speed	Slow (Newspapers/Radio)	Instant (Real-time)
Accuracy	May be outdated	Highly Accurate (API Based)
Accessibility	Limited to specific times	Available 24/7
Cost	High (Travel to Mandi)	Free (Mobile Data only)

7 CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

- The project **"KRUSHI MITRA: A SMART AGRICULTURE APP"** has been successfully designed and developed using the **Flutter framework** and **Dart programming language**. The application successfully achieves its goal of providing real-time weather updates and market prices (Mandi rates) to farmers on a single mobile platform.

  Through this project, we have learned how to integrate **REST APIs** and parse **JSON data** to create a dynamic mobile application. The app's simple user interface makes it accessible for rural users, helping them make informed decisions about their crops. This project proves that modern technology can be effectively used to solve traditional agricultural challenges, ultimately supporting the "Digital India" mission.

7.2 Future Scope

- While the current version of the Krushi Mitra app is fully functional, there is always room for improvement. The future enhancements for this project include:

  1. **Multilingual Support:** Adding support for local languages like **Marathi and Hindi** to help farmers who are not comfortable with English.
  2. **AI-based Crop Disease Detection:** Using Machine Learning (ML) to allow farmers to upload photos of infected crops and get instant solutions or pesticide recommendations.
  3. **Direct Farmer-to-Buyer Marketplace:** Developing a feature where farmers can list their produce and sell it directly to wholesalers, removing middlemen and increasing profits.

1. **Expert Consultation:** Adding a chat or video call feature to connect farmers with agricultural scientists or experts for personalized advice.

2. **Offline Access:** Implementing local database storage (like SQLite) so that farmers can view the last updated market rates even without an active internet connection.

3. **Government Scheme Alerts:** Notifying users about new government subsidies, schemes, and agricultural news.

8 APPENDIX

A. Core Source Code Snippets

- The following code represents the logic used in the **"KRUSHI MITRA"** application to fetch real-time weather data and handle the API responses.

1. Weather Fetching Logic (using OpenWeather API):

```
// API Integration for fetching city-wise weather data
final String myApiKey = "714e3f06973e6a9b793e031b6117d876";

Future<void> fetchWeather(String city) async {
  if (city.isEmpty) return;
  try {
    final url = "https://api.openweathermap.org/data/2.5/weather?q=$city,IN&appid=$myApiKey&units=metric&lang=mr";
    final res = await http.get(Uri.parse(url));

    if (res.statusCode == 200) {
      // Successfully received data from the server
      updateUI(jsonDecode(res.body));
    } else {
      // Handled city not found or invalid response
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text("शहर सापडले नाही!"))
      );
    }
  } catch (e) {
    debugPrint("Error: $e");
  }
}
```

A. User Manual (How to use the App)

-To ensure the ease of use for farmers, follow these steps:

1. **Step 1:** Install the `KrushiMitra.apk` on an Android device.
2. **Step 2:** Open the application and navigate to the "Weather" section.
3. **Step 3:** Enter the name of the city (e.g., Jintur) in the search bar.
4. **Step 4:** View real-time temperature, humidity, and weather descriptions in Marathi.
5. **Step 5:** Navigate to the "Markets Prics" section to see the latest crop prices for Soyabean, Cotton, etc.

B. List of Abbreviations

- **API:** Application Programming Interface
- **UI:** User Interface
- **JSON:** JavaScript Object Notation
- **SDK:** Software Development Kit
- **HTTP:** Hypertext Transfer Protocol
- **MSBTE:** Maharashtra State Board of Technical Education

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10. • **Capacitor Community (2023)** 'Cross-platform Native Runtimes for Web Apps', Capacitor Documentation. Available at: https://capacitorjs.com/docs.

PROJECT LINK :

Github Repo Link :

Table:6.3 Comparison with Manual Methods

1. **Flutter Documentation:** https://docs.flutter.dev/ - For UI development and framework guidelines.
2. **Dart Language Tour:** https://dart.dev/ - For implementing backend logic and functions.
3. **OpenWeather API:** https://openweathermap.org/api - For fetching real-time weather data.
4. **Google Gemini AI:** Used as an **AI Research & Design Assistant** for project documentation, technical logic, and generating the **Official Logo & Icons** for the "Krushi Mitra" application.
5.